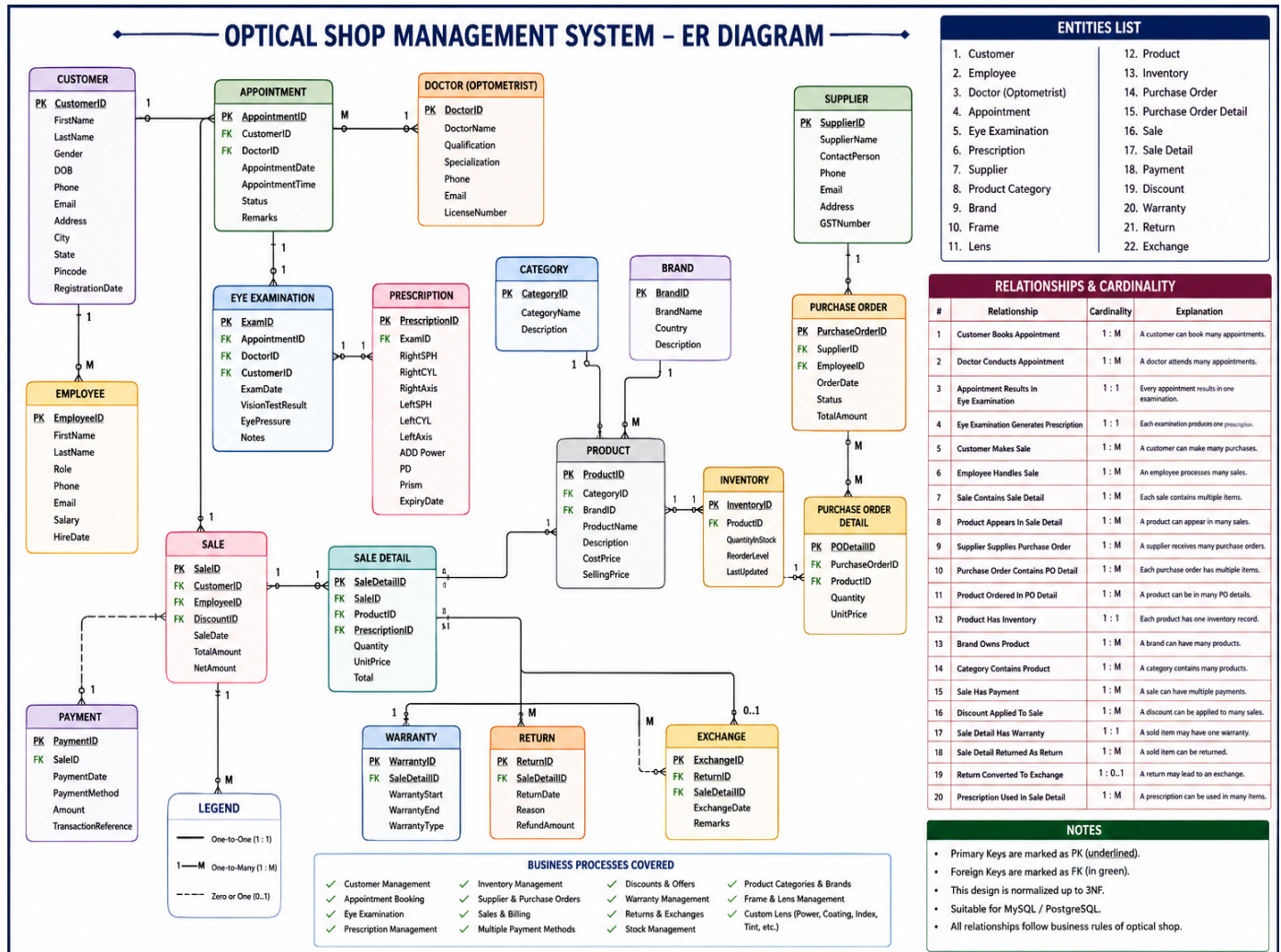


# Optical Shop Management System - ER Diagram Explanation

This document explains the Entity Relationship (ER) Diagram for an Optical Shop Management System. The database is normalized up to Third Normal Form (3NF) and supports customer management, appointments, eye examinations, prescriptions, inventory, purchasing, sales, payments, returns, exchanges, warranties, suppliers, and stock management.



## Entity Explanations

### Customer

Stores customer personal and contact information. One customer can book many appointments and make many sales.

### Appointment

Links customers with doctors for scheduled eye examinations.

## Doctor

Stores optometrist information. One doctor can conduct many appointments.

## Eye Examination

Contains examination results such as vision test, eye pressure and notes.

## Prescription

Stores lens power (SPH, CYL, Axis, ADD, PD, Prism) generated from an eye examination.

## Employee

Employees handle sales and purchase orders.

## Supplier

Suppliers provide products through purchase orders.

## Product, Category and Brand

Products are classified by category and brand. Frames and lenses belong to brands.

## Inventory

Tracks available stock and reorder levels for every product.

## Purchase Order & Details

Purchase orders contain multiple products purchased from suppliers.

## Sale & Sale Details

A sale belongs to one customer and contains multiple sale items.

## Payment

Supports multiple payment methods such as Cash, Card and UPI.

## Warranty

Warranty details are maintained for sold products.

## Return & Exchange

Returned products may be refunded or exchanged.

## Normalization

The design follows 1NF, 2NF and 3NF to eliminate redundancy and improve consistency.

## Relationship Summary

- Customer → Appointment (1:M)
- Doctor → Appointment (1:M)
- Appointment → Eye Examination (1:1)
- Eye Examination → Prescription (1:1)
- Customer → Sale (1:M)
- Sale → Sale Detail (1:M)
- Product → Sale Detail (1:M)
- Supplier → Purchase Order (1:M)
- Purchase Order → Purchase Order Detail (1:M)
- Product → Inventory (1:1)
- Sale → Payment (1:M)
- Sale Detail → Warranty (1:1)
- Sale Detail → Return (1:M)
- Return → Exchange (1:0..1)

## Conclusion

This ER model provides a scalable and efficient database design suitable for implementing a complete Optical Shop Management System using MySQL or PostgreSQL.